

Week 2 - Workshop

C# Variables and Datatypes

Task Type	Individual Work
Description	In today's workshop, we will write and run C# program containing variables and DataTypes concepts. • All the tasks are Individual. • Detail description of each tasks are mentioned in their respective questions slides. • All questions must be completed within workshop session. Total: 6, Pending: 1,2 Partial: 3, Completed: 4,5,6
Time Allocation	- [Teacher] Task Debrief: 20 min - [Student] Task 1: 5 min - [Student] Task 2: 5 min - [Student] Task 3: 5 min - [Student] Task 4: 10 min - [Student] Task 5: 15 min - [Student] Task 6: 15 min - [Teacher] Task Review: 15 Min
Submission of Work	Direct review by the teacher

Variables and DataTypes

Task 1: Write a C# program that performs the following: [5 Minutes]

- Declare a variable named **userName** to store your name and initialize it.
- Declare a variable named **luckyNumber** to store an integer and initialize it to your favorite single-digit number.
- Use **string interpolation** to print a single line of output to the console that says: "Hello, [userName]! Your lucky number is [luckyNumber] ."

- Task 1
- Task 2
- Task 3
- Task 4
- Task 5
- Task 6

Variables and DataTypes

Task 2: Constants [5 Minutes]

- Create a class named **Circle**.
- Inside the class, declare a **constant** variable named **PI** and initialize it with the value 3.14.
- In the **Main()** method of **Program.cs**,
 - Try to **modify the value of PI** of the Circle class and **observe the compilation error** and explain the error.
- Additionally, you can create methods to calculate area and perimeter of the circle.

- Task 1

- **Task 2**

- Task 3

- Task 4

- Task 5

- Task 6

Variables and DataTypes

Task 3: Data Types and Type Conversion [5 minutes]

- Declare and initialize variables of the following types: byte, short, int, long, float, double, decimal, char, bool.
- Convert the integer value 42 to a string and store it in a new variable.
- Convert a string "3.14" to a double and store it in a new variable.
- Print all variables to the console with appropriate labels showing their types and values.

- Task 1
- Task 2
- **Task 3**
- Task 4
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Variables and DataTypes

Task 4: Array and Array Methods [10 minutes]

- Create a single-dimensional integer array containing 5 of your favorite numbers.
- Use `Array.Sort()` to sort the array in ascending order.
- Use `Array.Reverse()` to reverse the sorted array.
- Print each element of the array using a for loop.
- Use `Array.IndexOf()` to find the position of a specific number in the array.

- Task 1
- Task 2
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Variables and DataTypes

Task 5: Date Time and Time Span [15 minutes]

- Create a DateTime variable representing your birthdate.
- Create another DateTime variable representing the current date and time.
- Calculate your age using TimeSpan (you'll need to subtract the two DateTime values).
- Print your birthdate, the current date, and your age in years.
- Add 10 days to your birthdate and prints the resulting date.

- Task 1
- Task 2
- Task 3
- Task 4
- **Task 5**
- Task 6

Variables and DataTypes

Task 6: Generic Collections (List and Dictionary) [15 minutes]

- Create a List<string> containing the names of 3 of your favorite fruits.
- Add a new fruit to the list.
- Remove one fruit from the list.
- Print all fruits in the list using a foreach loop.
- Create a Dictionary<int, string> where keys are fruit IDs (1,2,3) and values are the fruit names.
- Add a new entry to the dictionary and prints all key-value pairs.

- Task 1
- Task 2
- Task 3
- Task 4
- Task 5
- Task 6